# Homework #1

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## 1.

As shown in the below q-q plots, daily log-returns of all 4 tickers have heavier tails than the normal distribution, but as time interval increases to monthly, log-returns become more normal, consistent with the stylized facts. Except GME, all daily log-returns have small autocorrelations, although for some lags they are outside of the 95% confidence interval. GME has larger autocorrelations, which is against Rama Cont's observation. There is a gains/loss asymmetry for all tickers except PFE. For all tickers the log-returns clearly have volatility clustering, which is again consistent with Rama Cont's observation.

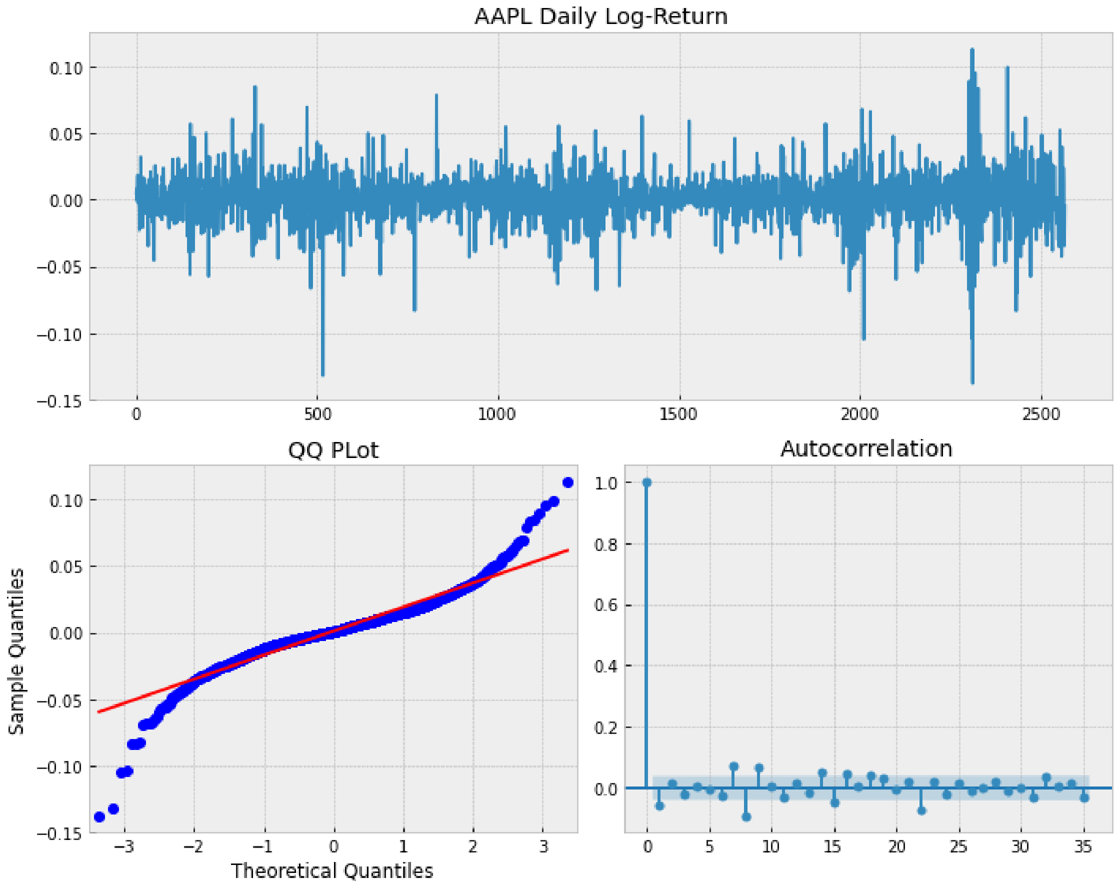
[2]:
```
import os
import sys
import pandas as pd
import pandas_datareader as pdr
import numpy as np
import statsmodels as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.api as smt
import statsmodels.api as sm
import scipy.stats as scs
import matplotlib.pyplot as plt
import matplotlib as mp
import seaborn as sns
```

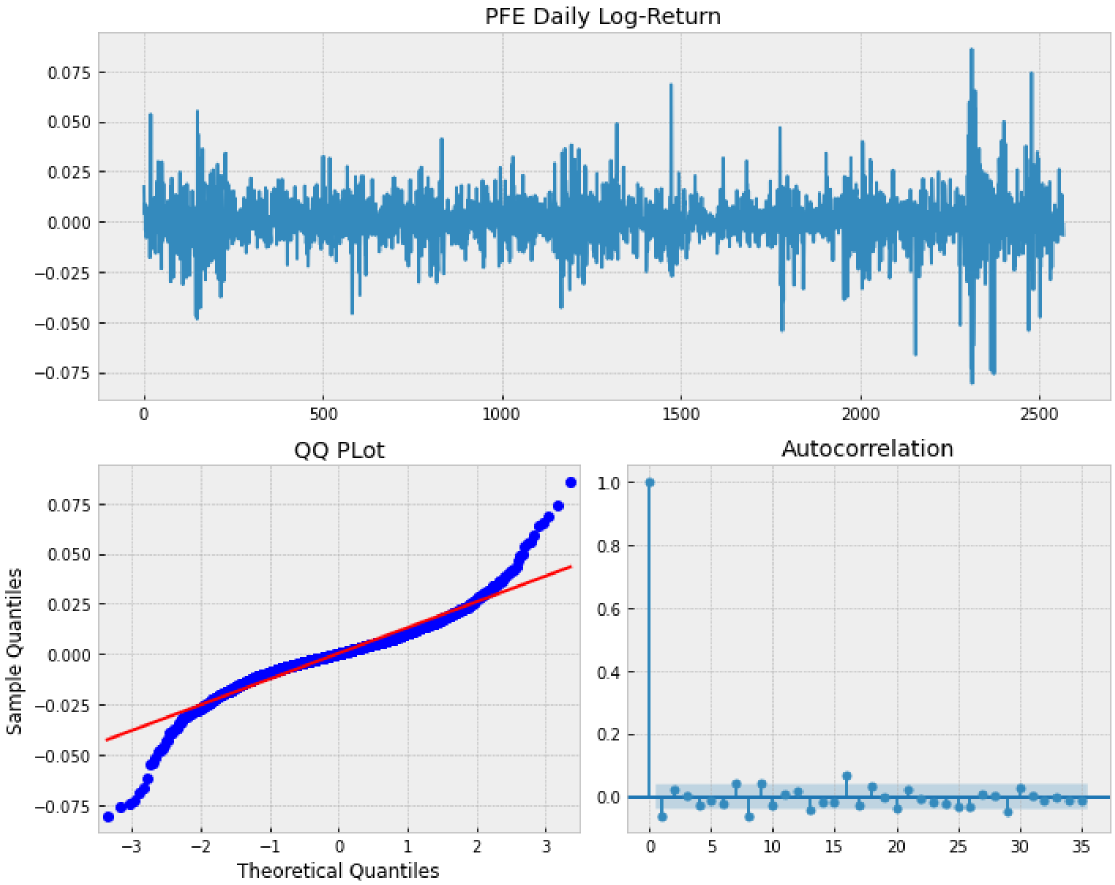
[3]:
```
def tsplot(y, lags=None, figsize=(10,8), style='bmh', title=None):
    if not isinstance(y, pd.Series):
        y = pd.Series(y)
    with plt.style.context(style):
        fig = plt.figure(figsize=figsize)
        layout = (2,2)
        ts_ax = plt.subplot2grid(layout, (0,0), colspan=2)
        acf_ax = plt.subplot2grid(layout, (1,1))
        qq_ax = plt.subplot2grid(layout, (1,0))
        y.plot(ax=ts_ax)
```

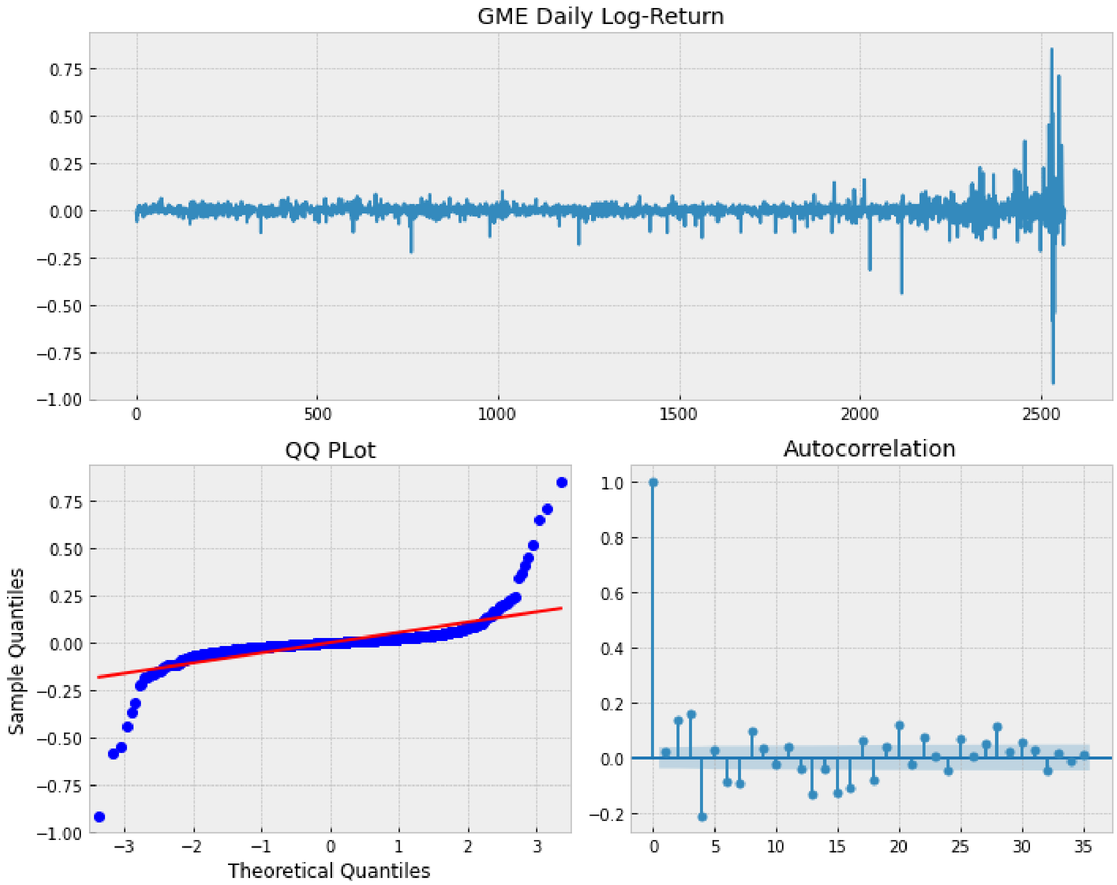
```
        ts_ax.set_title(title)
        smt.graphics.plot_acf(y, lags=lags, ax=acf_ax, alpha=0.05)
        sm.qqplot(y, line='s', ax=qq_ax)
        qq_ax.set_title('QQ PLot')
        plt.tight_layout()
    return
```

[4]:

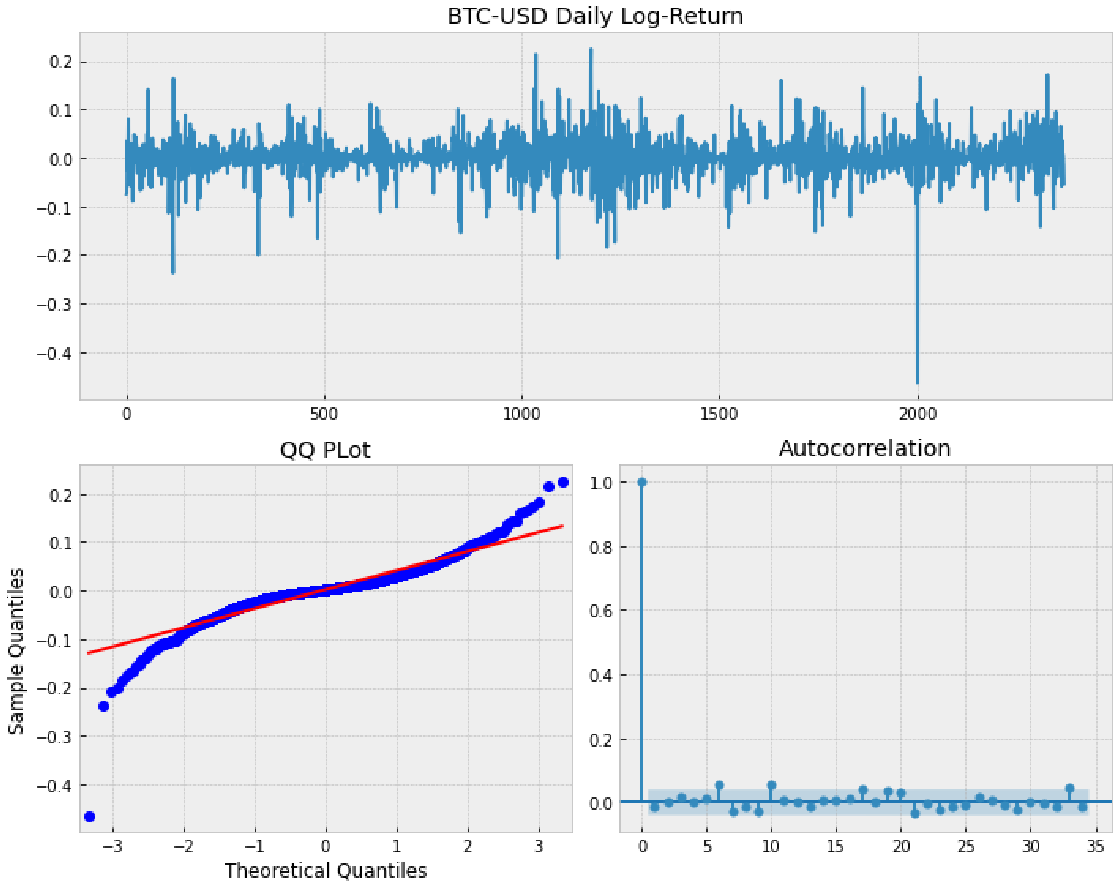
```
for ticker in ['aapl', 'pfe', 'gme', 'btc-usd']:
    start = '2011-01-01'
    end = '2021-03-21'
    price = pdr.get_data_yahoo(ticker,start,end)
    adjclose = price['Adj Close']
    shf = 1
    logret = np.log(adjclose/adjclose.shift(shf)).dropna()
    L = len(logret)
    LL = int(L/shf)
    logret_shf = np.zeros(LL)
    for i in range(int(L/shf)):
        logret_shf[i] = logret[i*shf]

    tsplot(logret_shf, title=f'{ticker.upper()} Daily Log-Return')
    plt.show()
```


AAPL Daily Log-Return
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation


PFE Daily Log-Return
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation

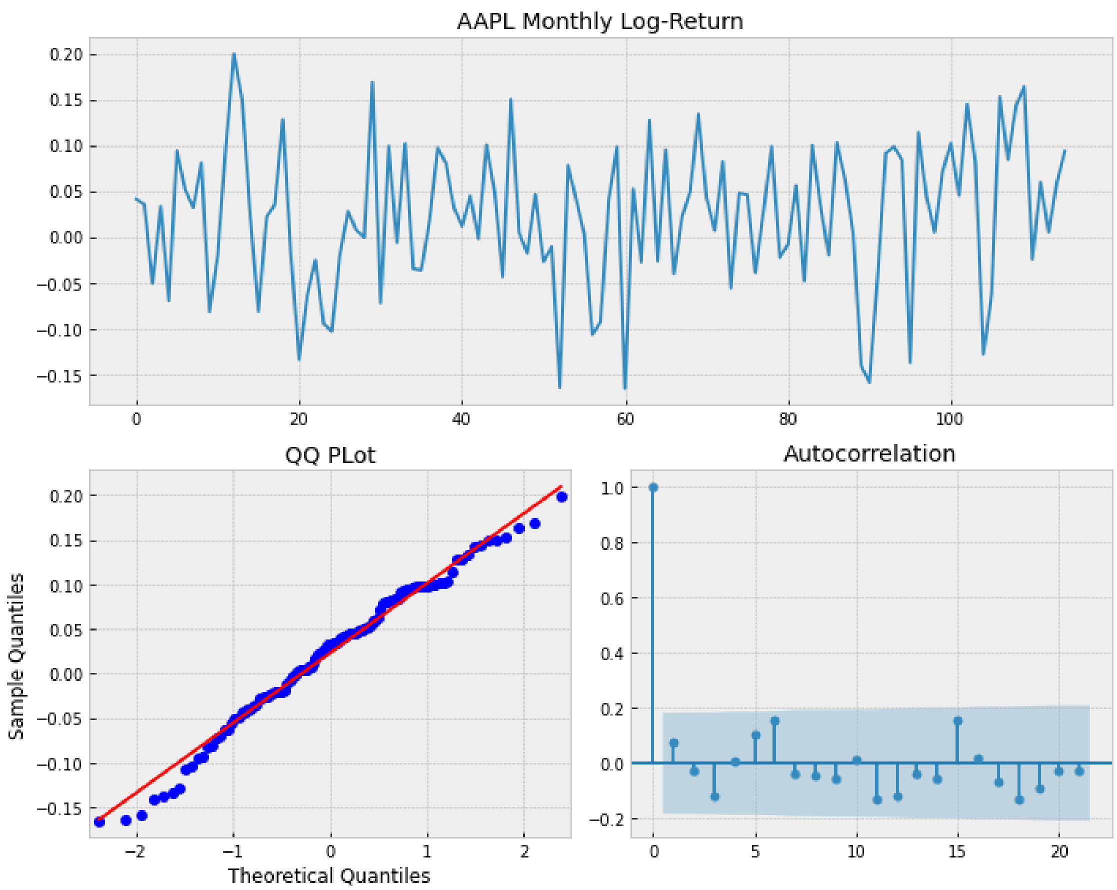


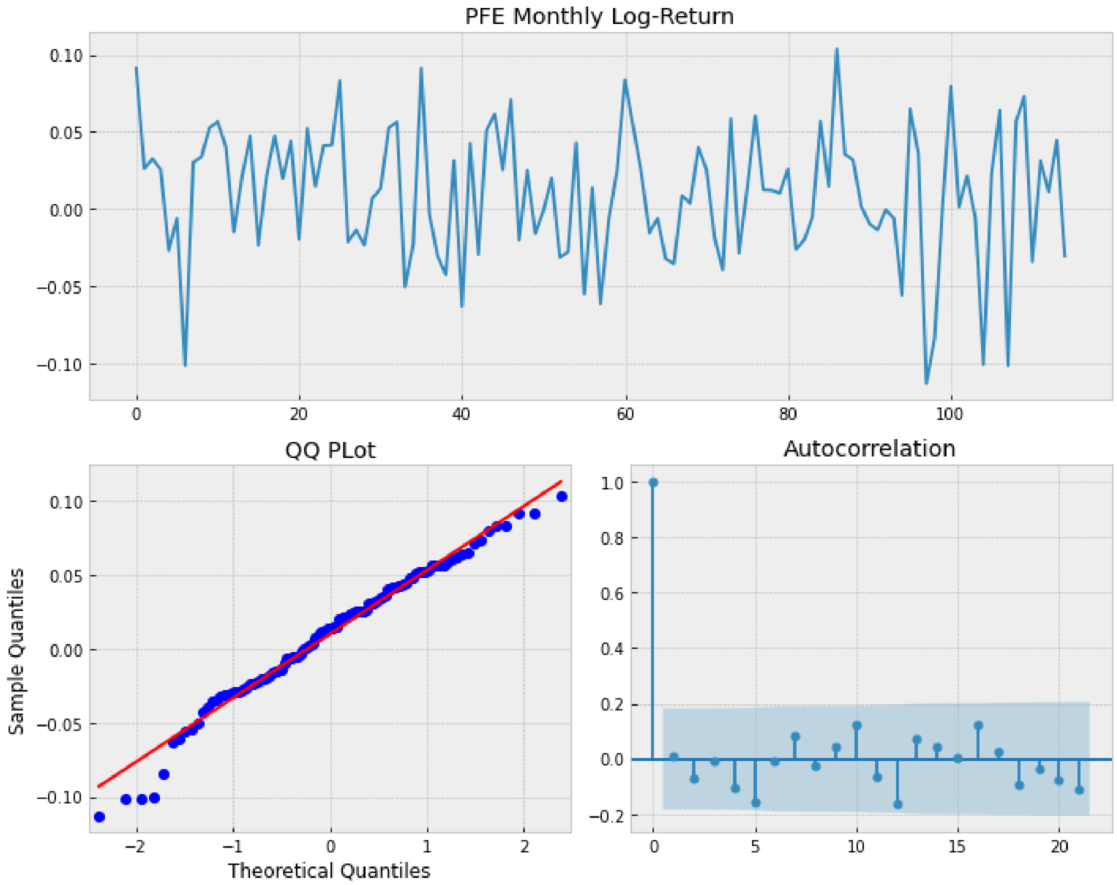
GME Daily Log-Return
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
500
1000
1500
2000
2500
QQ PLot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
Autocorrelation
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0
5
10
15
20
25
30
35

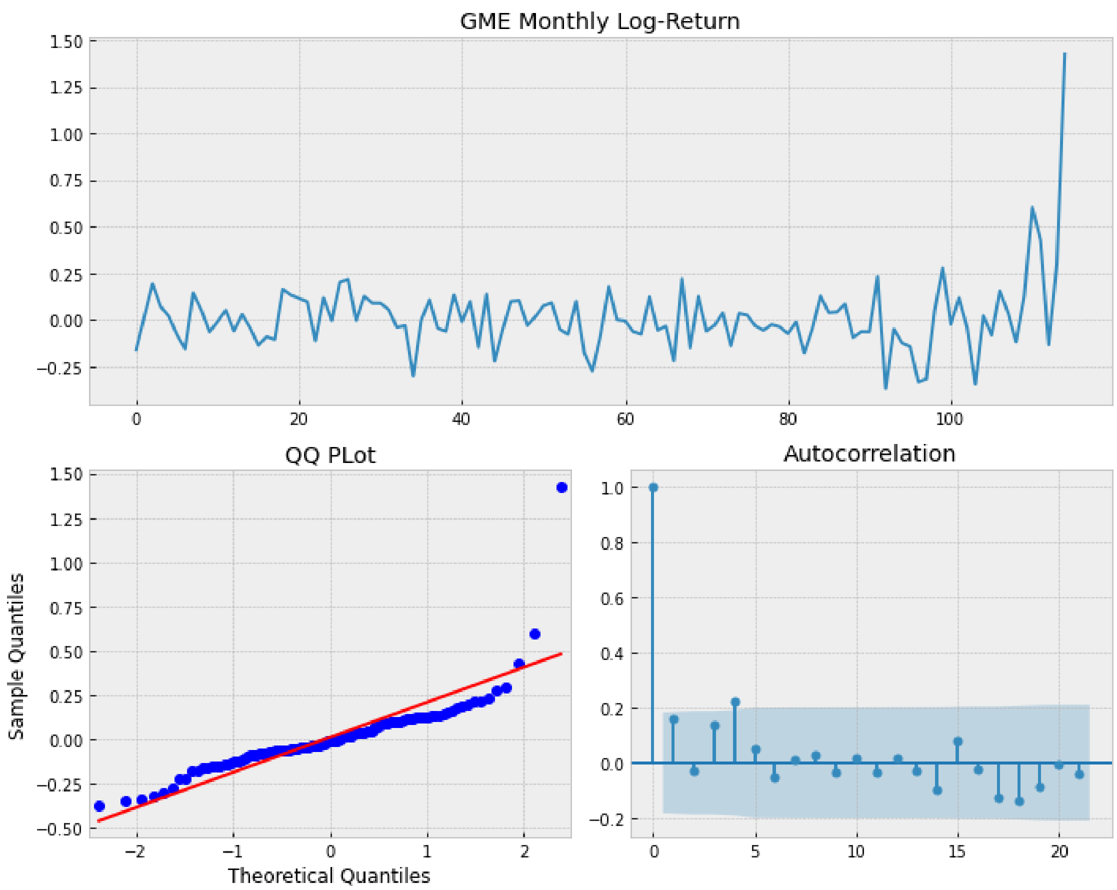


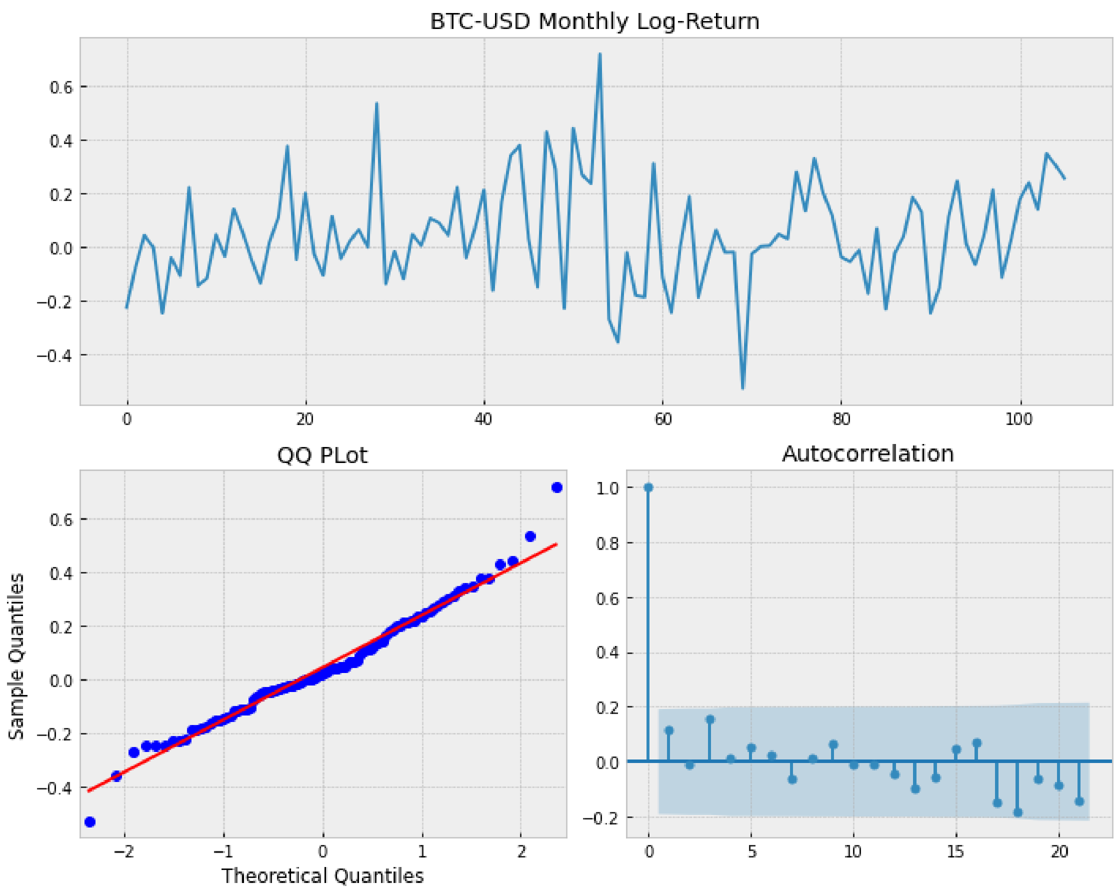

[5]:

```
for ticker in ['aapl', 'pfe', 'gme', 'btc-usd']:
    start = '2011-01-01'
    end = '2021-03-21'
    price = pdr.get_data_yahoo(ticker,start,end)
    adjclose = price['Adj Close']
    shf = 22
    logret = np.log(adjclose/adjclose.shift(shf)).dropna()
    L = len(logret)
    LL = int(L/shf)
    logret_shf = np.zeros(LL)
    for i in range(int(L/shf)):
        logret_shf[i] = logret[i*shf]

    tsplot(logret_shf, title=f'{ticker.upper()} Monthly Log-Return')
    plt.show()
```


AAPL Monthly Log-Return
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation


PFE Monthly Log-Return
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation


GME Monthly Log-Return
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation

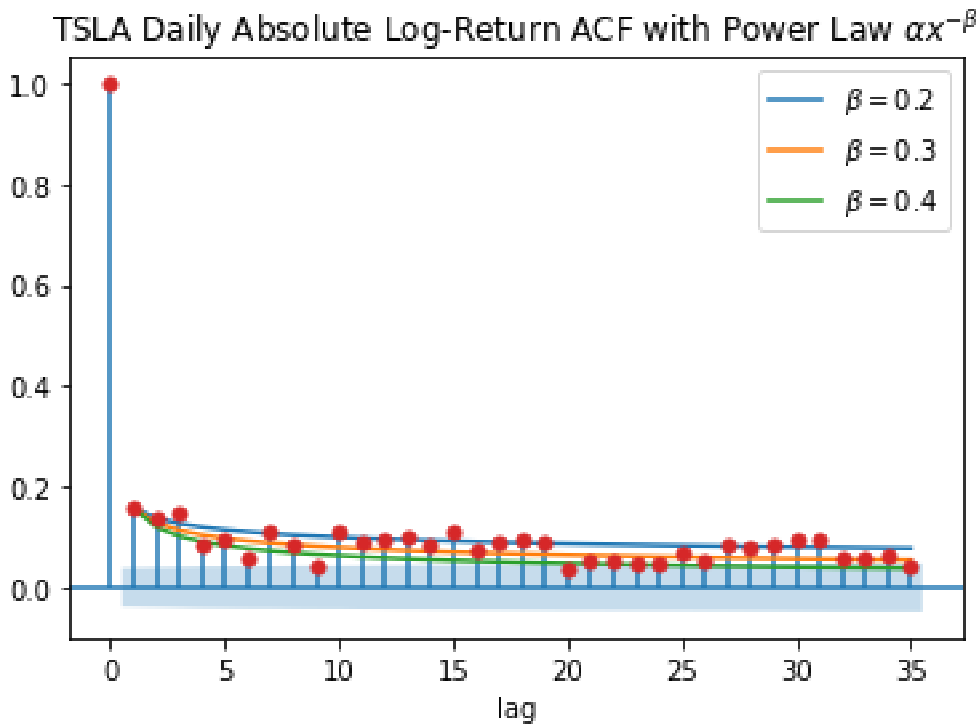


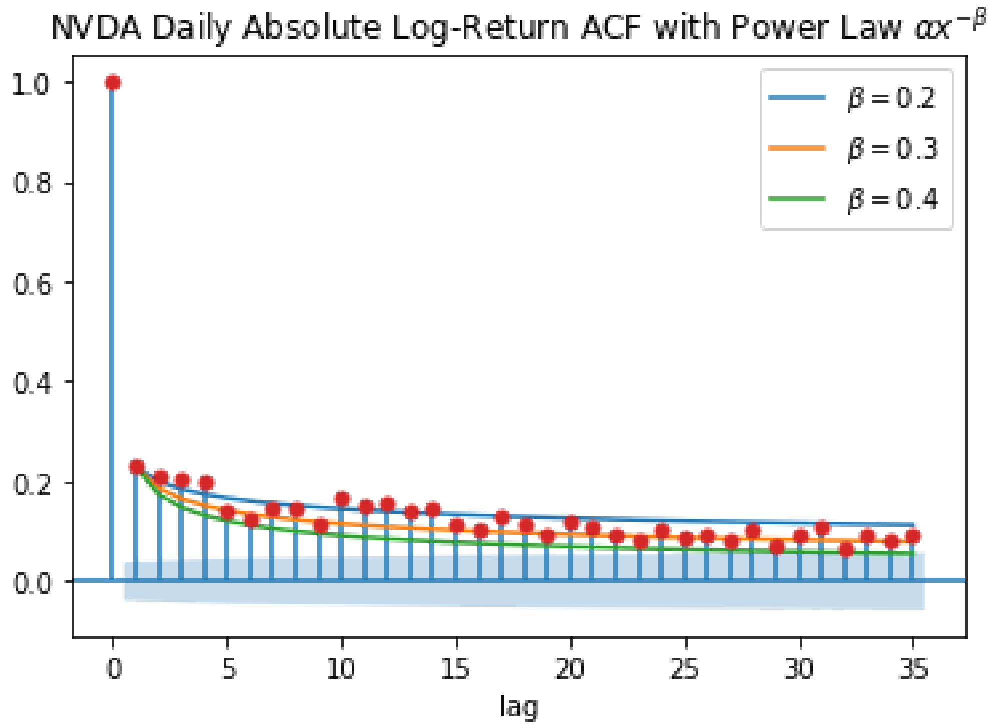

## 2.

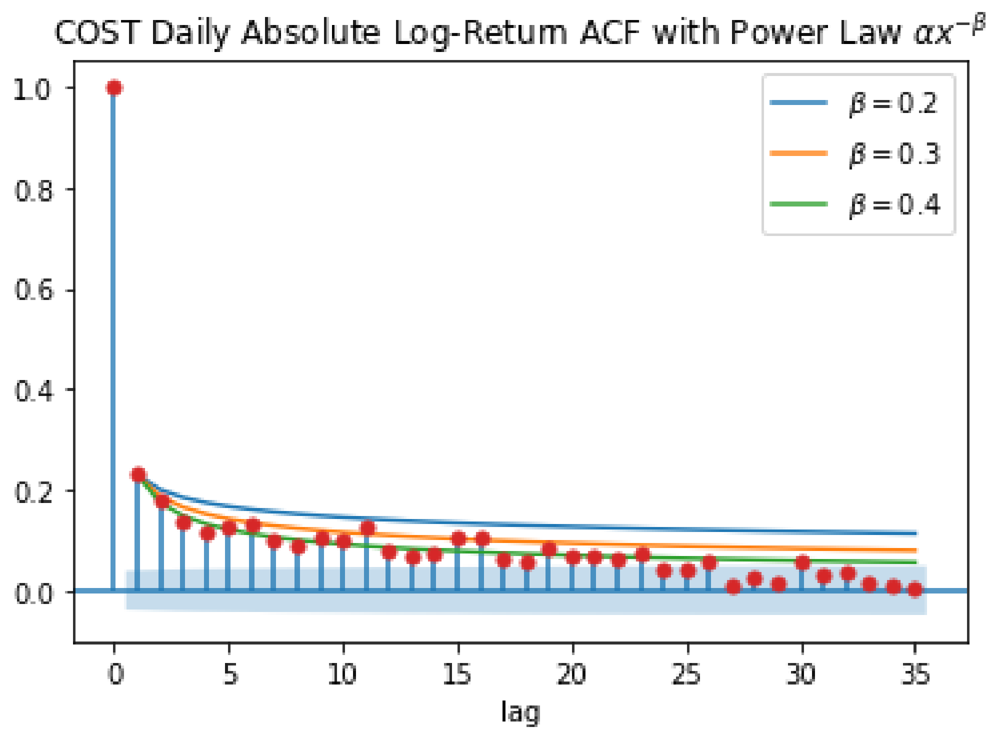
For Tesla (TSLA), Nvidia (NVDA) and Costco (COST), we plot the ACF of the absolute return and the power law $\alpha x^{-\beta}$ for $\beta = 0.2, 0.3$ and $0.4$. We match $\alpha$ and the autocorrelation at lag 1. Roughly speaking the ACFs follow the power law, with maybe an exception of Costco at long lags which has lower autocorrelation than the power low.

[20]:
```
import matplotlib.pyplot as plt
from pandas import DataFrame
from statsmodels.tsa.stattools import acf
import warnings
warnings.filterwarnings('ignore')

for ticker in ['tsla', 'nvda', 'cost']:
    start = '2011-01-01'
    end = '2021-03-21'
    price = pdr.get_data_yahoo(ticker,start,end)
    adjclose = price['Adj Close']
```

```
    shf = 1
    logret = np.log(adjclose/adjclose.shift(shf)).dropna()
    L = len(logret)
    LL = int(L/shf)
    logret_shf = np.zeros(LL)
    for i in range(int(L/shf)):
        logret_shf[i] = logret[i*shf]

    fig, ax = plt.subplots(1, 1)
    alpha = acf(np.abs(logret_shf))[1]
    DataFrame([(x, alpha*x**(-0.2), alpha*x**(-0.3), alpha*x**(-0.4)) for x in␣
↪range(1, 36)], columns=['lag', r'$\beta=0.2$', r'$\beta=0.3$', r'$\beta=0.
↪4$']).set_index('lag').plot(ax=ax)
    smt.graphics.plot_acf(np.abs(logret_shf), title=r'%s Daily Absolute␣
↪Log-Return ACF with Power Law $\alpha x^{-\beta}$'%ticker.upper(), alpha=0.
↪05, ax=ax)
```




NVDA Daily Absolute Log-Return ACF with Power Law $\alpha x^{-\beta}$
$\beta = 0.2$
$\beta = 0.3$
$\beta = 0.4$
lag


COST Daily Absolute Log-Return ACF with Power Law $\alpha x^{-\beta}$
$\beta = 0.2$
$\beta = 0.3$
$\beta = 0.4$
lag

## 3.

### (a)

First note that the mean is

$$E(X_t) = E(a_0 + a_1\epsilon_t + a_2\epsilon_{t-1}) = a_0.$$

Next we compute $\mathrm{Cov}(X_s, X_t)$. Without loss of generality, assume $s \leq t$. When $s = t$, we have

$$\begin{aligned}
\mathrm{Cov}(X_s, X_t) &= \mathrm{Cov}(X_t, X_t) \\
&= \mathrm{Var}(X_t) \\
&= \mathrm{Var}(a_0 + a_1\epsilon_{t-1} + a_2\epsilon_{t-2}) \\
&= (a_1^2 + a_2^2)\sigma^2.
\end{aligned}$$

When $s = t - 1$, we have

$$\begin{aligned}
\mathrm{Cov}(X_s, X_t) &= \mathrm{Cov}(X_{t-1}, X_t) \\
&= \mathrm{Cov}(a_0 + a_1\epsilon_{t-1} + a_2\epsilon_{t-2}, a_0 + a_1\epsilon_t + a_2\epsilon_{t-1}) \\
&= E[(a_0 + a_1\epsilon_{t-1} + a_2\epsilon_{t-2})(a_0 + a_1\epsilon_t + a_2\epsilon_{t-1})] - a_0^2 \\
&= a_1 a_2 \sigma^2 - a_0^2.
\end{aligned}$$

Here for all $t_i \neq t_j$, we have the cross term $E[\epsilon_{t_i}\epsilon_{t_j}] = E[\epsilon_{t_i}]E[\epsilon_{t_j}] = 0$. Thus the only nonzero term is $E[a_1 a_2 \epsilon_{t-1}^2] = a_1 a_2 \sigma^2$.

For the same reason, when $s \leq t - 2$, we will have $E[X_s X_t] = 0$ and hence $\mathrm{Cov}(X_s, X_t) = -a_0^2$.

From the above argument it is clear that $\mathrm{Cov}(X_s, X_t)$ only depends on the difference between $t$ and $s$, which remains unchanged when both are shifted by $k$ (meaning $(t + k) - (s + k) = t - s$). Thus we conclude that $X_t$ is weakly stationary with autocovariance function

$$\gamma(h) = \begin{cases} (a_1^2 + a_2^2)\sigma^2 & \text{if } h = 0 \\ a_1 a_2 \sigma^2 - a_0^2 & \text{if } h = 1 \\ 0 & \text{if } h \geq 2 \end{cases}.$$

### (b)

The mean is

$$E(X_t) = E(a_0 + a_1 f(\epsilon_t)) = a_0 + a_1 E(f(\epsilon_t)).$$

Since $\epsilon_t$ might have a different distribution for each $t$, so do $f(\epsilon_t)$. Thus the value of $E(f(\epsilon_t))$ might also be different for each $t$. For example, if $f(x) = x^3$ then $E(f(\epsilon_t))$ is the third moment of $\epsilon_t$ which is not guaranteed to be constant over time. Therefore the process is not weakly stationary.

### (c)

The mean is

$$E(X_t) = E\left(\prod_{i=0}^{k} \epsilon_{t-i}\right) = \prod_{i=0}^{k} E\left(\epsilon_{t-i}\right) = 0,$$

independent of $t$.

For $\mathrm{Cov}(X_s, X_t)$, note that when $s = t$, we have

$$\begin{aligned}
\mathrm{Cov}(X_s, X_t) &= \mathrm{Cov}(X_t, X_t) \\
&= E\left(\prod_{i=0}^{k} \epsilon_{t-i}^2\right) \\
&= \prod_{i=0}^{k} E\left(\epsilon_{t-i}^2\right) = \sigma^{2k}.
\end{aligned}$$

On the other hand, when $s < t$, we have

$$\begin{aligned}
\mathrm{Cov}(X_s, X_t) &= E\left[\left(\prod_{i=0}^{k} \epsilon_{s-i}\right)\left(\prod_{i=0}^{k} \epsilon_{t-i}\right)\right] \\
&= E\left[\left(\prod_{i=0}^{k} \epsilon_{s-i}\right)\left(\prod_{i=1}^{k} \epsilon_{t-i}\right)\epsilon_t\right].
\end{aligned}$$

Now since $s < t$, $\epsilon_t$ is not in the sequence $\{\epsilon_{s-i}\}_{i=0}^{k}$ nor $\{\epsilon_{t-i}\}_{i=1}^{k}$, and we can write

$$\mathrm{Cov}(X_s, X_t) = E\left[\left(\prod_{i=0}^{k} \epsilon_{s-i}\right)\left(\prod_{i=1}^{k} \epsilon_{t-i}\right)\right] E(\epsilon_t) = 0.$$

Thus $\mathrm{Cov}(X_s, X_t)$ only depends on the lag $s - t$ (in fact it only depends on if the lag is 0). We conclude that $X_t$ is weakly stationary with autocovariance function

$$\gamma(h) = \begin{cases} \sigma^{2k} & \text{if } h = 0 \\ 0 & \text{otherwise} \end{cases}.$$

## 4.

Assuming $\epsilon_t$ are also independent sequence, we have the mean

$$E(X_t) = E(\epsilon_t + \epsilon_{t-1}\epsilon_{t-2}) = E(\epsilon_t) + E(\epsilon_{t-1})E(\epsilon_{t-2}) = 0$$

and the covariance

$$\begin{aligned}
\mathrm{Cov}(X_t, X_{t+h}) &= E[(\epsilon_t + \epsilon_{t-1}\epsilon_{t-2})(\epsilon_{t+h} + \epsilon_{t+h-1}\epsilon_{t+h-2})] \\
&= E[\epsilon_t\epsilon_{t+h}] + E[\epsilon_{t-1}\epsilon_{t-2}\epsilon_{t+h}] + E[\epsilon_t\epsilon_{t+h-1}\epsilon_{t+h-2}] + E[\epsilon_{t-1}\epsilon_{t-2}\epsilon_{t+h-1}\epsilon_{t+h-2}].
\end{aligned}$$

When $h = 0$, given the independence assumption, the second and the third expectation is 0 and the covariance reduces to

$$\text{Cov}(X_t, X_t) = E[\epsilon_t^2] + E[\epsilon_{t-1}^2]E[\epsilon_{t-2}^2] = \sigma^2 + \sigma^4.$$

When $h > 0$, the 4 expectations $E[\epsilon_t \epsilon_{t+h}]$, $E[\epsilon_{t-1}\epsilon_{t-2}\epsilon_{t+h}]$, $E[\epsilon_t \epsilon_{t+h-1}\epsilon_{t+h-2}]$ and $E[\epsilon_{t-1}\epsilon_{t-2}\epsilon_{t+h-1}\epsilon_{t+h-2}]$ are all zeros because of the independence assumption. Thus $X_t$ is a white noise $\text{WN}(0, \sigma^2 + \sigma^4)$.

To see if $X_t$ is an MDS, note that

$$\begin{aligned} E[X_{t+1}|\mathcal{F}_t] &= E[\epsilon_{t+1} + \epsilon_t \epsilon_{t-1}|\mathcal{F}_t] \\ &= E[\epsilon_{t+1}|\mathcal{F}_t] + \epsilon_t \epsilon_{t-1} = \epsilon_t \epsilon_{t-1}. \end{aligned}$$

Given the filtration at time $t$, $\epsilon_t$ and $\epsilon_{t-1}$ are realizations, not random variables, which may or may not be zero. So $X_t$ is not an MDS.